ISE 3200 - HW1: Modeling & the Graphical Method

Ethan Brooks

Q1. Buckeye Furniture makes tables and chairs. A table needs 4 hours of carpentry and 2 hours of finishing; a chair needs 3 hours of carpentry and 1 hour of finishing. Each week 240 carpentry hours and 100 finishing hours are available. Each table earns $70 profit and each chair $50. Formulate an LP that maximizes weekly profit.

Q1a) Define the decision variables (with units).

x1 = tables built per week, x2 = chairs built per week

Q1b) Write the objective function.

max z = 70x1 + 50x2

Q1c) Write all constraints.

4x1 + 2x2 <= 240; 3x1 + x2 <= 100; x1, x2 >= 0

Q2. Solve the LP from Q1 graphically.

Q2a) List every corner point of the feasible region.

(0, 0), (50, 0), (0, 80)

Q2b) Give the optimal solution and its profit.

x1 = 30, x2 = 40, z = $4,100

Q3) Suppose the finishing constraint is removed. Is the LP still bounded? Give the new optimum.

Still bounded: carpentry limits both products. New optimum x1 = 60, x2 = 0, z = $4,200.